

Ayudantía 9- MAT1620

1. Aplique la regla de la cadena para encontrar $\frac{dz}{dt}$

(a) $z = x^2 + y^2 + xy, x = \sin(t), y = e^t.$

(b) $z = xe^{y/z}, x = t^2, z = 1 + 2t.$

2. Aplique la regla de la cadena para encontrar $\frac{\partial z}{\partial s}$ y $\frac{\partial z}{\partial t}$

(a) $z = \arcsin(x - y), x = s^2 + t^2, y = 1 - 2st.$

(b) $e^r \cos(\theta), r = st, \theta = \sqrt{s^2 + t^2}.$

3. Suponga que f es una función derivable en variables x e y , y que

$$g(u, v) = f(e^u + \sin(v), e^u + \cos(v))$$

Use la tabla de valores para calcular $g_u(0, 0)$ y $g_v(0, 0)$.

	f	g	f_x	f_y
$(0, 0)$	3	6	4	8
$(1, 2)$	6	3	2	5

4. Demuestre que cualquier función de la forma

$$z = f(x + at) + g(x - at)$$

es una solución de la ecuación diferencial

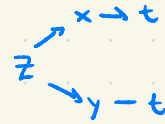
$$\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

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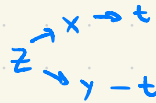


$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$\frac{dz}{dt} = (2x + y) \cdot \cos(t) + (2y + x)e^t$$

$$\Rightarrow \frac{dz}{dt} = (2\sin(t) + e^t) \cos(t) + (2e^t + \sin(t))e^t$$

b) $z = xe^{y/z}, x = t^2, y = 1 + 2t$



$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

↳ $F(x, y, z) = z - xe^{y/z} = 0$

↳ $F_x(x, y, z) = -e^{y/z}$

↳ $F_y(x, y, z) = -\frac{x}{z} e^{y/z}$

↳ $F_z(x, y, z) = 1 - x \cdot e^{y/z} \cdot \left(-\frac{y}{z^2}\right) = 1 + \frac{xy}{z^2} e^{y/z} \neq 0$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{e^{y/z}}{1 + \frac{xy}{z^2} e^{y/z}}$$

$$\Rightarrow \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = \frac{\frac{x}{z} e^{y/z}}{1 + \frac{xy}{z^2} e^{y/z}}$$

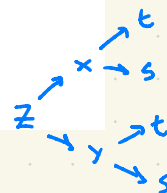
$$\Rightarrow \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$\Rightarrow \frac{dz}{dt} = \frac{e^{y/z}}{1 + \frac{xy}{z^2} e^{y/z}} \cdot 2t + \frac{\frac{x}{z} e^{y/z}}{1 + \frac{xy}{z^2} e^{y/z}} \cdot 2$$

2. Aplique la regla de la cadena para encontrar $\frac{\partial z}{\partial s}$ y $\frac{\partial z}{\partial t}$

(a) $z = \arcsen(x - y), x = s^2 + t^2, y = 1 - 2st.$

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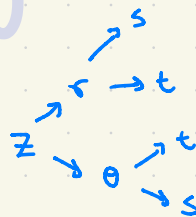
a) $z = \arcsin(x - y), x = s^2 + t^2, y = 1 - 2st$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial s} = \frac{1}{\sqrt{1 - (x - y)^2}} \cdot 2s + \frac{-1}{\sqrt{1 - (x - y)^2}} \cdot -2t$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$\frac{\partial z}{\partial t} = \frac{1}{\sqrt{1 - (x - y)^2}} \cdot 2t + \frac{-1}{\sqrt{1 - (x - y)^2}} \cdot -2s$$



b) $z = e^r \cos(\theta), r = st, \theta = \sqrt{s^2 + t^2}$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial r} \cdot \frac{\partial r}{\partial s} + \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial s}$$

$$\frac{\partial z}{\partial s} = e^r \cos(\theta) \cdot t + -e^r \sin(\theta) \cdot \frac{1}{2\sqrt{s^2 + t^2}} \cdot 2s$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial r} \cdot \frac{\partial r}{\partial t} + \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial t}$$

$$\frac{\partial z}{\partial t} = e^s \cos(\theta) \cdot s + -e^s \sin(\theta) \cdot \frac{t}{\sqrt{s^2 + t^2}}$$

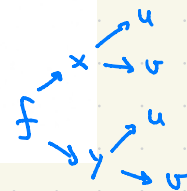
3. Suponga que f es una función derivable en variables x e y , y que

$$g(u, v) = f(e^u + \sin(v), e^u + \cos(v)) = f(x, y)$$

Use la tabla de valores para calcular $g_u(0, 0)$ y $g_v(0, 0)$.

	f	g	f_x	f_y
$(0, 0)$	3	6	4	8
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$$g(u, v) = f(x, y)$$



$$x = e^u + \sin(v), \quad y = e^u + \cos(v)$$

$$\frac{\partial g}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$g_u = f_x \cdot e^u + f_y \cdot e^u$$

$$g_u(u, v) = f_x(x, y) \cdot e^u + f_y(x, y) \cdot e^u$$

$$g_u(u, v) = f_x(e^u + \sin(v), e^u + \cos(v)) \cdot e^u + f_y(e^u + \sin(v), e^u + \cos(v)) \cdot e^u$$

$$g_u(0, 0) = f_x(1, 2) \cdot 1 + f_y(1, 2) \cdot 1$$

$$\Rightarrow g_u(0, 0) = f_x(1, 2) + f_y(1, 2)$$

$$\Rightarrow g_u(0, 0) = 2 + 5 = 7$$

$$\frac{\partial g}{\partial v} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$g(u, v) = f(x, y)$$

$$\frac{\partial g}{\partial u} = \frac{\partial f(x, y)}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \dots$$

$$\frac{\partial g}{\partial v} = f_x \cdot \cos(v) + f_y \cdot -\sin(v)$$

$$\Rightarrow g_v(u, v) = f_x(x, y) \cdot \cos(v) + f_y(x, y) \cdot -\sin(v)$$

$$\Rightarrow g_v(0, 0) = f_x(1, 2) \cdot 1 + f_y(1, 2) \cdot 0$$

$$\Rightarrow g_v(0, 0) = 2$$

4. Demuestre que cualquier función de la forma

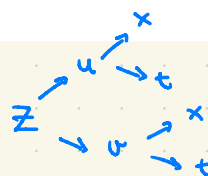
$$z = f(x + at) + g(x - at), \quad a \in \mathbb{R}$$

es una solución de la ecuación diferencial

$$\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

$$u = x + at, \quad v = x - at$$

$$z(u, v) = f(u) + g(v)$$



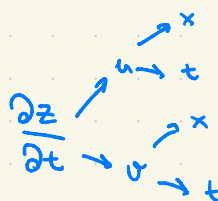
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial t}$$

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial u} + \cancel{\frac{\partial g}{\partial u}}$$

$$\frac{\partial z}{\partial v} = \cancel{\frac{\partial f}{\partial v}} + \frac{\partial g}{\partial v}$$

$$\Rightarrow \frac{\partial z}{\partial t} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial t}$$

$$\Rightarrow \frac{\partial z}{\partial t} = f_u \cdot a - g_v \cdot a$$



$$\Rightarrow \frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial u} (f_u \cdot a - g_v \cdot a) \cdot \frac{\partial u}{\partial t} + \frac{\partial}{\partial v} (f_u \cdot a - g_v \cdot a) \cdot \frac{\partial v}{\partial t}$$

$$\frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial u} (f_u \cdot a - g_v \cdot a) \cdot \frac{\partial u}{\partial t} + \frac{\partial}{\partial v} (f_u \cdot a - g_v \cdot a) \cdot \frac{\partial v}{\partial t}$$

$$\frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial u} (f_u \cdot a) \cdot \frac{\partial u}{\partial t} + \frac{\partial}{\partial v} (-g_v \cdot a) \cdot \frac{\partial v}{\partial t}$$

$$\Rightarrow \frac{\partial^2 z}{\partial t^2} = a \cdot f_{uu} \cdot a + -g_{vv} \cdot a \cdot -a$$

$$\Rightarrow \frac{\partial^2 z}{\partial t^2} = a^2 f_{uu} + a^2 \cdot g_{vv} = a^2 (f_{uu} + g_{vv})$$

$\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 g}{\partial v^2}$

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot 1 + \frac{\partial g}{\partial v} \cdot 1 = f_u + g_v$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial u} (f_u + g_v) \cdot \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} (f_u + g_v) \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial^2 z}{\partial x^2} = f_{uu} \cdot 1 + g_{vv} \cdot 1 = f_{uu} + g_{vv}$$

$$\Rightarrow \frac{\partial^2 z}{\partial t^2} = a^2 \cdot \frac{\partial^2 z}{\partial x^2}$$